

# KnowServe Corp. — Engineering Department

## Technical Debt Reduction Plan

### Goal

Reduce cumulative technical debt by 30 % by Q4 2025 to improve system reliability and developer velocity.

### Debt Assessment

Quarterly scans (using SonarQube) produce a Tech Debt Ratio = Remediation Cost / Development Cost.

Current ratio = 0.42 (42 %). Target < 0.30 by December 2025.

### Priority Categories

1. High – Security flaws and deprecated libraries (30 % of backlog) .
2. Medium – Inefficient database queries and duplicated logic .(% 45)
3. Low – Stylistic inconsistencies and missing documentation.(% 25)

### Action Plan

- Refactor 2 legacy microservices per quarter .
- Replace obsolete API gateways by Q3 .
- Automate static analysis in CI/CD to prevent regressions .
- Allocate 15 % of engineering capacity each sprint to tech-debt tasks.

### Measurement

KPIs tracked monthly: build time, bug recurrence rate, test coverage improvement.

A Tech-Debt Dashboard in Grafana visualizes progress company-wide.